

Time evolution operator

We seek a time-evolution operator $U(t, t_0)$ such that

$$\psi(x, t) = U(t, t_0)\psi(x, t_0)$$

Start with the Schrodinger equation.

$$i\hbar \frac{\partial}{\partial t} \psi(x, t) = H(t)\psi(x, t)$$

Substitute for $\psi(x, t)$.

$$i\hbar \frac{\partial}{\partial t} [U(t, t_0)\psi(x, t_0)] = H(t)U(t, t_0)\psi(x, t_0)$$

Expand the left-hand side by the product rule.

$$i\hbar \frac{\partial}{\partial t} U(t, t_0)\psi(x, t_0) + \boxed{i\hbar U(t, t_0) \frac{\partial}{\partial t} \psi(x, t_0)} = H(t)U(t, t_0)\psi(x, t_0)$$

The boxed term drops out due to $\partial\psi(x, t_0)/\partial t = 0$.

$$i\hbar \frac{\partial}{\partial t} U(t, t_0)\psi(x, t_0) = H(t)U(t, t_0)\psi(x, t_0)$$

By operator equivalence

$$i\hbar \frac{\partial}{\partial t} U(t, t_0) = H(t)U(t, t_0)$$

Rearrange as

$$\frac{1}{U(t, t_0)} \partial U(t, t_0) = -\frac{i}{\hbar} H(t) \partial t$$

Integrate both sides.

$$\int_{t_0}^t \frac{1}{U(\tau, t_0)} dU(\tau, t_0) = -\frac{i}{\hbar} \int_{t_0}^t H(\tau) d\tau$$

Solve the first integral.

$$\log U(t, t_0) - \boxed{\log U(t_0, t_0)} = -\frac{i}{\hbar} \int_{t_0}^t H(\tau) d\tau$$

By $U(t_0, t_0) = 1$ the boxed term vanishes.

$$\log U(t, t_0) = -\frac{i}{\hbar} \int_{t_0}^t H(\tau) d\tau$$

Exponential of both sides.

$$U(t, t_0) = \exp \left(-\frac{i}{\hbar} \int_{t_0}^t H(\tau) d\tau \right) \tag{1}$$

For time-independent H the integral can be replaced with $(t - t_0)H$.

$$U(t, t_0) = \exp \left(-\frac{i}{\hbar} H \int_{t_0}^t d\tau \right) = \exp \left(-\frac{i}{\hbar} (t - t_0) H \right)$$

Eigenmath script